

```
for (let i=1; i<=3; i++){
  log("Hello")
}
```

initialization

condition

logic

Update

$i=1$

$i<=3$

$\log("Hello")$

$i=i+1$

$i=1$

$1<=3 (T)$

Hello

$i=1+1=2$

$i=2$

$2<=3 (T)$

Hello

$i=2+1=3$

$i=3$

$3<=3 (T)$

Hello

$i=3+1=4$

$i=4$

$4<=3 (F)$

X

X

even number in the range of 1 to 10

```
for (let i=1; i<=10; i++){
```

```
  if ( $i\%2===0$ ){
```

```
    console.log(i)
```

```
  }
```

```
}
```

initialization $i=1$	Condition $i \leq 10$	logic if $i \% 2 == 0$ $dg(i)$	updatation $i = i + 1$
$i=1$	$1 \leq 10$ (T)	$1 \% 2 == 0$ (F)	$i=2$
$i=2$	$2 \leq 10$ (T)	$2 \% 2 == 0$ (T) 2	$i=3$
$i=3$	$3 \leq 10$ (T)	$3 \% 2 == 0$ (F)	$i=4$
$i=4$	$4 \leq 10$ (T)	$4 \% 2 == 0$ (T) 4	$i=5$
$i=5$	$5 \leq 10$ (T)	$5 \% 2 == 0$ (F)	$i=6$
$i=6$	$6 \leq 10$ (T)	$6 \% 2 == 0$ (T) 6	$i=7$
$i=7$	$7 \leq 10$ (T)	$7 \% 2 == 0$ (F)	$i=8$
$i=8$	$8 \leq 10$ (T)	$8 \% 2 == 0$ (T) 8	$i=9$
$i=9$	$9 \leq 10$ (T)	$9 \% 2 == 0$ (F)	$i=10$
$i=10$	$10 \leq 10$ (T)	$10 \% 2 == 0$ (T) 10	$i=11$
$i=11$	$11 \leq 10$ (F)	(T) $\rightarrow$ X	X

Print odd numbers from 1 to 5

```
for (let i=1; i<=5; i++) {
  if (i%2!=0) {
    dg(i)
  }
}
```

initialisation $i=1$	condition $i \leq 5$	logic (code condition if True print(i)	updatation $i=i+1$
$i=1$	$1 \leq 5$ (T)	$1 \% 2 != 0$ (T) 1	$i=2$
$i=2$	$2 \leq 5$ (T)	$2 \% 2 != 0$ (F)	$i=3$
$i=3$	$3 \leq 5$ (T)	$3 \% 2 != 0$ (T) 3	$i=4$

$i=4$

$4 \leq 5 (T)$

$4 \times 2! = 0 (F)$

$i=5$

$i=5$

$5 \leq 5 (T)$

$5 \times 2! = 0 (T)$

$i=6$

$i=6$

$6 \leq 5 (F)$

5

X

X

factorial of a given number using for loop.

$\text{let } n=4$

$\text{let fact}=1$

for (let i=1; i≤n; i++){

$\text{fact} = \text{fact} * i;$

}

console.log(fact)

Initialisation

Condition

Logic/Code

Update

$i=1$

$i \leq 4$

$\text{fact} = \text{fact} * i$

$i = i + 1$

$i=1$

$1 \leq 4 (T)$

$\text{fact} = 1 * 1$
 $= 1$

$i = 1 + 1 = 2$

$i=2$

$2 \leq 4 (T)$

$\text{fact} = 1 * 2$
 $= 2$

$i = 3$

$i=3$

$3 \leq 4 (T)$

$\text{fact} = 2 * 3$
 $= 6$

$i = 4$

$i=4$

$4 \leq 4 (T)$

$\text{fact} = 6 * 4$

$i = 5$

$i=5$

$5 \leq 4 (F)$

$\text{fact} = 24$
X

X